Physics - Undergraduate Level Assessment 1 Total Marks: 30 Answer all questions.

Multiple Choice (4 marks)

1. A single-electron atom has the electron in the l = 2 state. The number of allowed values of the quantum number m_l is

   (a) 5

   (b) 4

   (c) 3

   (d) 2

2. A grating spectrometer can just barely resolve two wavelengths of 500 nm and 502 nm, respectively. Which of the following gives the resolving power of the...

   (a) 2

   (b) 250

   (c) 5,000

   (d) 10,000

True / False (26 marks)

Answer True or False and justify briefly.

1. True or False: The correct answer to 'The emission spectrum of the doubly ionized lithium atom Li++ (Z = 3, A = 7) is identical to that of a hydrogen atom in which all the...' is 'decreased by a factor of 9'.

2. True or False: The correct answer to 'An organ pipe, closed at one end and open at the other, is designed to have a fundamental frequency of C (131 Hz). What is the frequency...' is '262 Hz'.

3. True or False: The correct answer to 'If the total energy of a particle of mass m is equal to twice its rest energy, then the magnitude of the particle's relativistic momentum is' is '(3^(1/2))mc'.

4. True or False: The correct answer to 'Electromagnetic radiation emitted from a nucleus is most likely to be in the form of' is 'gamma rays'.

5. True or False: The correct answer to 'A particle decays in 2.0 ms in its rest frame. If the same particle moves at v=0.60c in the lab frame, how far will it travel in the lab...' is '450 m'.

6. True or False: The correct answer to 'For which of the following thermodynamic processes is the increase in the internal energy of an ideal gas equal to the heat added to the...' is 'Constant volume'.

7. True or False: The correct answer to 'The negative muon, mu^-, has properties most similar to which of the following?' is 'Electron'.

8. True or False: The correct answer to 'An atom has filled n = 1 and n = 2 levels. How many electrons does the atom have?' is '10'.

9. True or False: The correct answer to 'A three-dimensional harmonic oscillator is in thermal equilibrium with a temperature reservoir at temperature T. The average total energy...' is '(1/2) k T'.

10. True or False: The correct answer to 'De Broglie hypothesized that the linear momentum and wavelength of a free massive particle are related by which of the following constants?' is 'Planck's constant'.

11. True or False: The correct answer to 'The eigenvalues of a Hermitian operator are always' is 'real'.

12. True or False: The correct answer to 'According to the BCS theory, the attraction between Cooper pairs in a superconductor is due to' is 'the weak nuclear force'.

13. True or False: The correct answer to 'The speed of light inside of a nonmagnetic dielectric material with a dielectric constant of 4.0 is' is '1.5 x 10^8 m/s'.

End of Paper